

Quiz (Habanero)

- Q1.** What is the factored form of $x^3 + 8$?
(A) $(x + 2)(x^2 - 2x + 4)$
(B) $(x + 2)(x^2 + 2x + 4)$
(C) $(x + 2)(x^2 - 2x - 4)$
(D) $(x - 2)(x^2 + 2x + 4)$
(E) $(x - 2)(x^2 - 2x - 4)$
- Q2.** A glacier melts at a rate of $x^3 - 27$ cubic meters per hour. What is the factored form?
(A) $(x - 3)(x^2 + 3x + 9)$
(B) $(x - 3)(x^2 - 3x + 9)$
(C) $(x + 3)(x^2 - 3x + 9)$
(D) $(x + 3)(x^2 + 3x - 9)$
(E) $(x - 3)(x^2 + 3x - 9)$
- Q3.** A forest's temperature changes by $x^3 + 64$ degrees per day. Factor this expression.
(A) $(x + 4)(x^2 - 4x + 16)$
(B) $(x + 4)(x^2 + 4x + 16)$
(C) $(x - 4)(x^2 - 4x - 16)$
(D) $(x - 4)(x^2 + 4x - 16)$
(E) $(x + 4)(x^2 - 4x - 16)$
- Q4.** What is the factored form of $27x^3 - 8y^3$?
(A) $(3x - 2y)(9x^2 + 6xy + 4y^2)$
(B) $(3x + 2y)(9x^2 - 6xy + 4y^2)$
(C) $(3x - 2y)(9x^2 - 6xy - 4y^2)$
(D) $(3x + 2y)(9x^2 + 6xy - 4y^2)$
(E) $(3x + 2y)(9x^2 + 6xy + 4y^2)$
- Q5.** A hurricane's wind speed increases by $x^3 - 125$ miles per hour. Factor this expression.
(A) $(x - 5)(x^2 + 5x + 25)$
(B) $(x - 5)(x^2 - 5x + 25)$
(C) $(x + 5)(x^2 + 5x - 25)$
(D) $(x + 5)(x^2 - 5x + 25)$
(E) $(x + 5)(x^2 + 5x + 25)$
- Q6.** What is the factored form of $64x^3 + 125y^3$?
(A) $(4x + 5y)(16x^2 - 20xy + 25y^2)$
(B) $(4x - 5y)(16x^2 + 20xy + 25y^2)$
(C) $(4x + 5y)(16x^2 + 20xy + 25y^2)$
(D) $(4x + 5y)(16x^2 - 20xy - 25y^2)$
(E) $(4x - 5y)(16x^2 - 20xy + 25y^2)$
- Q7.** A weather balloon rises at a rate of $x^3 + 216$ feet per minute. Factor this expression.
(A) $(x + 6)(x^2 - 6x + 36)$
(B) $(x + 6)(x^2 + 6x + 36)$
(C) $(x - 6)(x^2 - 6x + 36)$
(D) $(x - 6)(x^2 + 6x + 36)$
(E) $(x + 6)(x^2 + 6x - 36)$
- Q8.** A drought's effect on a lake's water level changes by $x^3 - 1000$ feet per year. What is the factored form?
(A) $(x - 10)(x^2 + 10x + 100)$
(B) $(x - 10)(x^2 - 10x + 100)$
(C) $(x - 10)(x^2 + 10x - 100)$
(D) $(x + 10)(x^2 + 10x - 100)$
(E) $(x + 10)(x^2 + 10x + 100)$

Open Ended Questions (FRQ)

- Q9.** Factor $x^3 + 343y^3$ and explain the pattern used for factoring.
- Q10.** Factor $27x^3 - 64y^3$ and describe the similarities between this problem and a real-world environmental scenario.

Chef's Tips

- Review the difference of cubes formula $(a^3 - b^3) = (a - b)(a^2 + ab + b^2)$
- Have students research real-world examples of weather patterns and environmental changes
- Provide extra support for students who struggle with factoring complex expressions